

Dev Khullar

(he/him) | [@devkhullar](#) | d.khullar@uleth.ca
website: devkhullar.github.io

Education

B.Sc. Mathematics

University of Lethbridge, Lethbridge, AB, Canada

September 2023 - April 2027

3.80/4.00 GPA

Research Experience

Astrophysics Research Assistant

April 2025 - Present

Supervisor: Drs. Qiana Hunt and Alex Tetarenko

University of Lethbridge, Lethbridge, AB, Canada

- Performed large-scale, multiwavelength surveys of X-ray binaries in nearby young, spiral galaxies using the Chandra X-ray Observatory, the Hubble Space Telescope and the James Webb Space Telescope.
- Used this survey to calculate the ejection velocities of X-ray binaries out of their parent clusters. [[scientific analysis](#)]
- Developed a semi-automated photometric workflow for aperture photometry and mass estimation of stars with the James Webb Space Telescope. This code is now helping Dr. Hunt in their [primary research](#).
- Currently **preparing a first-author publication on our results** with key [PHANGS](#) collaborators that is to be submitted to the Astrophysical Journal Letters.

Machine Learning Research Assistant

September 2025 - December 2025

Supervisor: Dr. Corina Birghila

University of Lethbridge, Lethbridge, AB, Canada

- Implemented reinforcement learning algorithms from research papers and from the text *Reinforcement Learning: An Introduction* by Barto and Sutton (2018) to test their limitations. [[code](#)]
- Presented my research findings at the Reinforcement Learning seminar at ULethbridge.
- This code is now being used to set up a reinforcement learning based hackathon by the Dept. of Computer Science at ULethbridge.

Mathematics Research Assistant

October 2024 - December 2024

Supervisor: Dr. Sean Fitzpatrick

University of Lethbridge, Lethbridge, AB, Canada

- Worked on showing the generalization of Stokes' Theorem using the language of differential forms and differential geometry.
- Presented applications of differential forms to Maxwell's Equations to extend my knowledge of differential geometry to theoretical physics. [[report](#)]

Astrophysics Research Assistant

May 2024 - September 2024

Supervisor: Dr. Alex Tetarenko

University of Lethbridge, Lethbridge, AB, Canada

- Analyzed archival and new radio observations of an X-ray binary in the core of the globular cluster Terzan 1 using the Very Large Array. [[scientific analysis](#)] [[report](#)]
- Used astronomical softwares CASA and CARTA for data processing, analysis and visualization of these observations to study the environment of this globular cluster.
- **Contributions from this research are going to be included in a future peer-reviewed paper that I will co-author.**

Employment

University Tutor September 2024 - April 2025
Accessible Learning Centre, University of Lethbridge, Lethbridge, AB, Canada

- Provided tutoring lessons to undergraduate students with disabilities in Linear Algebra, Calculus I & II, and Introduction to Computer Science.

Statistics Teaching Assistant January 2025 - April 2025
University of Lethbridge, Lethbridge, AB, Canada

- Undergraduate marker for Statistics 1770 - Introduction to Probability and Statistics.

Scholarships, Awards & Recognitions

As of December 2025, I have collectively received ~**CA\$26,850** in competitive awards and scholarships. Additionally, I have been on the Dean's Honour List in 4/5 completed semesters at ULethbridge. I was also honoured with the recognition of a 2024/2025 Shining Student, which recognizes exceptional students at the University of Lethbridge for their academic achievements and community involvement.

Dean's Honour List	Fall 2023, Winter 2024, Fall 2024 & Fall 2025
CASCA Funding Support (CA\$385)	2026
Myrtle Ruth Bodie Mathematics Scholarship (CA\$1000 × 2)	2025, 2026
University of Lethbridge Scholarship (CA\$600)	2025
Chinook Summer Research Award (CA\$6000 × 3)	2024, 2025
Student's Union Quality Initiatives Scholarship (CA\$1000 × 2)	2024, 2025
ULethbridge Shining Student	2024
International Continuing Award (CA\$1000)	2024
International Student Memorial Scholarship (CA\$700)	2024
International Entrance Award (CA\$1000)	2023
Board of Governors' Admission Scholarship (CA\$1000)	2023

Poster Presentations

<i>University of Lethbridge Summer Research Showcase</i>	August 2025
<i>Astrophysics Undergraduate Research Mini-Symposium</i>	September 2024
<i>University of Lethbridge Summer Research Showcase</i>	August 2024

Open Source Software

[*XRBIID*](#) – a python library for X-ray binary identification with the Chandra X-ray Observatory, the Hubble Space Telescope and the James Webb Space Telescope.

[*StarFinder*](#) – an open source game that I developed for outreach where the player has to detect the optimal amount of stars in an astronomical image. This game has been displayed at multiple events at ULethbridge, including high school visits and ULethbridge Open House (See Outreach).

[*rltoolbox*](#) – code implementation of simple reinforcement learning algorithms from textbooks and research papers.

[*euler_lagrange_solver*](#) – python code that symbolically solves the Euler-Lagrange equations for any coordinate.

Skills

Software: CASA | CARTA | SAOImage DS9 | Git/GitHub

Programming Languages: (*Expert*) Python, C++ | (*Intermediate*) R, html, CSS, JavaScript | (*Beg.*) Java

Outreach

Selected notable outreach contributions from the past year:

University of Lethbridge Observatory Team Member

October 2024 - Present

- Member of a team working on building an observatory at the University of Lethbridge for local and university outreach along with use for undergraduate classes.
- Secured a ULethbridge Arts and Science grant worth CA\$5,000 and purchased 3 telescopes for this project.
- I crowdsourced a CELESTRON NexStar 4SE telescope from the general public that is now operational for outreach

University of Lethbridge Open House Volunteer

October 2024 & 2025

- Facilitated lab tours of the Physics faculty premises, highlighting the various ongoing cutting-edge research happening in the Dept. of Physics and Astronomy to prospective students and their families.
- My open source game [StarFinder](#) was also presented at this event to highlight the work done by ULethbridge students.

ULethbridge Rebranding Volunteer

July 2025

- Represented the Department of Physics and Astronomy for ULethbridge's recent rebranding [see [video](#)] along with showing the cutting-edge telescope equipment available at ULethbridge.

InnovateHER Workshop, ULethbridge

July 2025

- Presented a workshop to middle school students on using Python for finding stars in galaxies (point-source identification) using the example of the galaxy IC 5332, along with investigating the galaxy with the visualization tool DS9. [[resources](#)]
- Secured MacBooks from ULethbridge for all the participants.

Physics and Astronomy Workshop Series Presenter, ULethbridge

July 2025

- Presented a workshop on advanced Python programming concepts (object oriented programming, debugging and software engineering) to fellow Physics faculty and students. [[Recording](#) (Passcode: H0A%Q8RC)][[resources](#)]
- Led a workshop on using astronomical visualization tool SAOImage DS9 and performing point source identification in astronomical images of galaxies in the Physics and Astronomy Department at ULethbridge. [[resources](#)]

Vera C. Rubin Observatory Watch Party Volunteer

June 2025

- Facilitated the organization and management of Alberta's only official watch party for the Vera C. Rubin Observatory's First look.

University of Lethbridge Math Fair Coordinator

February 2024, 2025 & 2026, December 2025

Pacific Institute for the Mathematical Sciences (PIMS)

- Engaged with the local Lethbridge community through various interactive Mathematics based games, events and puzzles (e.g. Fractals, Hanoi Tower, etc) for K-12 students, aimed at fostering their interest in Mathematics.

Media

Shining Student: I was interviewed by the University of Lethbridge for my academic achievements and community involvement.